quantmsdiann: a scalable SDRF-driven DIA-NN workflow for reanalysis of single-cell, spatial, and bulk proteomics datasets

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Abstract

Public proteomics archives now host thousands of data-independent acquisition (DIA) datasets, but they remain difficult to integrate and reuse: each was processed with a different software configuration, and most lack standardised metadata. Here, we present `quantmsdiann`, an open-source Nextflow/nf-

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core workflow that parallelises DIA-NN across cloud and HPC infrastructures, guided by the experimental design declared in SDRF format. The workflow ships container-pinned profiles for each supported DIA-NN version (extensible to any release via a custom container) through Docker and Singularity, with no manual installation or dependency resolution, and natively handles all vendor formats, converting to mzML when needed (e.g. SCIEX). It exports harmonised quantification tables as MSstats input, the new Quantitative Proteomics eXchange format (QPX), and a pmultiqc quality-control report. The parallelisation design reanalyses a 2,300-run single-cell dataset in 2.4 hours on an HPC cluster of 300 nodes. We benchmark quantmsdiann on the full ProteoBench DIA panel (five DIA-NN versions $\times$ four instrument modalities) and reanalyse public DIA deposits spanning bulk cell lines (NCI-60/PXD003539, ProCan/PXD030304), a single-cell oocyte plexDIA dataset (MSV000093870), a spatial Deep Visual Proteomics cohort (PXD064049), and a phosphoproteomics cohort (PXD049692). SDRF-driven reprocessing with current DIA-NN versions matches or exceeds the originally deposited bulk and single-cell results, recovers comparable precursor-level depth on the spatial cohort, and matches a directDIA phosphopeptide analysis; five cell-line cohorts are further combined into a pan-cohort under one invocation. quantmsdiann is a step towards scalable, reproducible reanalysis of the growing DIA data landscape and a foundation for community efforts in large-scale DIA meta-analysis and atlas building. quantmsdiann is available at https://github.com/bigbio/quantmsdiann.

Keywords: DIA-NN, single-cell proteomics, spatial proteomics, phosphoproteomics, plexDIA, data-independent acquisition, Nextflow, nf-core, SDRF, reproducibility, QPX, pmultiqc

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1. Introduction

Data-independent acquisition (DIA) is now the dominant mass-spectrometry strategy for proteomics at scale, and DIA-NN [1] is among the most widely adopted analysis engines across timsTOF [2, 3], Astral, and Orbitrap platforms. Recent benchmarks confirm DIA-NN's competitiveness for single-cell DIA workflows [4], and recent releases have expanded support for plexDIA, timsTOF parallel accumulation-serial fragmentation (PASEF), and vendor-native raw-file reading. Despite this analytical maturity, most DIA-NN users run the tool as a desktop application, which limits its scalability for large datasets. Deployments on HPC and cloud infrastructure remain artisanal, relying on shell scripts, ad-hoc directory conventions, and hand-curated sample sheets. As a result, the same raw data analysed by two laboratories, or with two `DIA-NN` releases, can yield non-overlapping quantification tables that are difficult to reconcile.

Beyond per-study analysis, the value of public proteomics archives such as PRIDE Archive [5] and other ProteomeXchange partners [6] lies as much in systematic reanalysis as in original deposition. In 2024 we published bigbio/quantms [7], a Nextflow pipeline that standardised multi-engine DIA and DDA analyses behind a Sample and Data Relationship Format (SDRF) [8] input on the nf-core framework [9]. quantms was, however, built around `DIA-NN` 1.8.1 with a fixed parameter set, and its DDA/DIA breadth made extending DIA-specific functionality difficult. New `DIA-NN` parameters conflicted with DDA options, versions were hard-coded, sample-level acquisition parameters were not properly propagated to per-run `DIA-NN` invocations, and raw-file conversion was mandatory. Iterative reanalysis at archive scale requires a workflow that combines the reproducibility of nf-core, the sensitivity of current `DIA-NN` releases, the metadata discipline of SDRF, and a parallelisation strategy capable of processing thousands of raw files.

Here, we present `quantmsdiann`, an independent SDRF-driven Nextflow pipeline dedicated to `DIA-NN` that extends DIA-specific features beyond what the original bigbio/quantms pipeline could accommodate. While `quantmsdiann` shares its SDRF-first design philosophy and parts of the configuration-generation tooling with quantms, its workflow, container set, and module library are developed and released independently, with container-pinned profiles for every `DIA-NN` version selectable at runtime under Docker or Singularity. Downstream, it exports a harmonised QPX (Quantitative Proteomics eXchange) Parquet archive alongside the standard `DIA-NN` reports and MSstats input, bundling the SDRF, identifications, quantifications, and pipeline provenance into a single queryable artefact for reanalysis and long-term archival. We demonstrate the pipeline along three axes: ProteoBench [10] benchmarks across four DIA modalities and five `DIA-NN` releases; independent reanalyses of public DIA deposits spanning bulk cell lines (NCI-60, ProCan), a single-cell oocyte plexDIA dataset, a spatial Deep Visual Proteomics cohort, and a phosphoproteomics cohort; and a pan-cohort of five public DIA cell-line reanalyses integrated under a single SDRF-driven invocation.

2. Experimental Procedures

2.1. Pipeline implementation

`quantmsdiann` is implemented in Nextflow DSL2 ($\geq$25.10.4) and follows nf-core community standards [9] for pipeline structure, testing, documentation, and continuous integration. It supports both `DIA-NN` library-free analysis (predicting the spectral library in-silico from a FASTA reference) and library-based analysis against a user-supplied spectral library; all analyses in this study used the library-free mode with the UniProt human reference proteome [11] (UP000005640, downloaded March 2026). Each computational step is packaged as a versioned container image (Docker, Singularity, or Apptainer). Open-source community tools (ThermoRawFileParser [12], pridepy [13], QPX, and the `wiffconverter` SCIEX reader) are pulled from public registries (BioContainers [14] or `ghcr.io/bigbio`). Because the `DIA-NN` license restricts redistribution, only version 1.8.1 is shipped as an image (the default, from BioContainers); releases from 1.9 onward (1.9.2–2.5.0) are built locally from the per-release recipes in the companion `quantms-containers` repository (`https://github.com/bigbio/quantms-containers`) and tagged so the workflow selects them automatically (Additional file 1). The repository at `https://github.com/bigbio/quantmsdiann` is organised as a standard nf-core pipeline with main workflow (`main.nf`), subworkflows, modules, test data and CI configurations, and documentation. Pipeline version v2.1.0 (release codename "Sao Pablo", 2026-05-08) was used for the analyses reported in this paper.

2.2. SDRF parsing and configuration generation

`quantmsdiann` accepts SDRF [8] as primary input. SDRF parsing is performed using the sdrf-pipelines Python package developed within the quantms project, which validates the SDRF against the proteomics-specific subset of EFO, PSI-MS, and PRIDE ontology terms and emits a normalised per-sample channel for downstream Nextflow processes. `DIA-NN` configuration generation extracts enzyme, fixed and variable modification, instrument, precursor mass tolerance, fragment mass tolerance, and missed-cleavage settings from the SDRF and writes a `DIA-NN` command-line configuration. Where SDRF columns are absent, `DIA-NN` defaults appropriate for the inferred instrument are applied.

2.3. Raw file handling and DIA-NN execution

Vendor formats are first-class inputs in `quantmsdiann`: unlike quantms, which converts every raw file to mzML, `quantmsdiann` reads most formats natively and converts only when required. Thermo `.raw` files are read natively by `DIA-NN` $\geq$ 2.1.0 (which added Linux native-reader support), or otherwise converted to mzML with ThermoRawFileParser [12] (`-mzml_convert`); Bruker `.d` folders are ingested natively, with compressed forms (`.d.tar`/`.zip`/`.gz`) decompressed first; and Sciex `.wiff` / `.wiff2` are converted to mzML with the open-source `wiffconverter` (`https://github.com/bigbio/wiffconverter`). Generic `.dia` and pre-converted `.mzML` inputs pass through unchanged, with optional OpenMS re-indexing (`-reindex_mzml`).

2.4. Quality control and QPX export

Per-run and per-cohort quality control reports are generated by `pmultiqc` [15], a proteomics extension of MultiQC [16] that summarises `DIA-NN`-specific metrics including peptide and protein counts, missed cleavages, charge-state distributions, precursor and fragment mass accuracy, and per-run identification yield. The QPX archive is produced by the `qpxc` CLI from the `qpx` package (`https://github.com/bigbio/qpx`), which writes Parquet files for identifications (psm, peptide, protein-group, feature), metadata (sample, run, ontology), and provenance, and optionally exports MuData (`.h5mu`) containers for scverse-based [17] downstream analysis. The aim of exporting to QPX is to bundle the SDRF, identifications, quantifications, and pipeline provenance into a single queryable artefact for reanalysis and long-term archival, and to provide a standardised input for downstream tools that support the format (e.g. MSstats, Perseus, or custom scripts).

2.5. Datasets benchmarked and compute environment

Additional file 1: Table S1 lists all benchmark and reanalysis datasets with their instrument, acquisition mode, and deposition year: the ProteoBench DIA modules 5/7/9/10, the three per-cohort cell-line reanalyses (PXD003539, PXD004701, PXD030304), the two additional pan-cohort datasets (PXD041421, PXD017199), the PXD064049 spatial Deep Visual Proteomics cohort, and the PXD071075 scalability sweep; it also provides each dataset's SDRF prepared from the original sample metadata. For each cell-line reanalysis cohort, the original published quantification was downloaded from PRIDE as the comparator, with both it and the `quantmsdiann` reanalysis thresholded at 1% precursor-level FDR; the conservative contaminant/decoy filter and the like-for-like-threshold rule applied to the original headline counts are described in Additional file 1. Runtime benchmarks were run on the EMBL-EBI HPC cluster (LSF scheduler), with per-task CPU and memory dispatched by Nextflow and memory profiled from the Nextflow trace report (`-with-trace`).

3. Results

3.1. `quantmsdiann` pipeline architecture

Built on the nf-core framework [9], `quantmsdiann` accepts a single SDRF file as primary input, which encodes per-sample metadata (cell line or tissue origin, factor values, instrument, post-translational modifications) and per-run file references. We extended the SDRF with DIA-relevant columns (MS1 and MS2 mass windows, and plexDIA channel definitions and labels) and added an adapter to sdrf-pipelines [8] (`convert-diann`) that translates each run's declared metadata into a per-run `DIA-NN` command-line configuration. Every run is therefore parametrised from its own metadata with no manual configuration or parameter-harmonisation step, removing a common source of analysis-script divergence between groups: different MS1/MS2 window schemes can be mixed within a cohort, or a plexDIA dataset run with its channel definitions, without any workflow change beyond the SDRF. The workflow comprises four mandatory and three optional modules organised around the `DIA-NN` analysis core (Fig. 1a), alternating between per-file parallel stages (red; raw-file preparation, preliminary calibration, and individual search) and collective stages (green; in-silico library generation, empirical library assembly, final quantification, MSstats conversion, and `pmultiqc` [15] quality-control reporting) operating on the full cohort.

`quantmsdiann` follows `DIA-NN`'s distributed multi-pass flow, alternating parallel per-file work with serial cohort-wide steps (Fig. 1a). Each raw file is first prepared *in parallel* (vendor-format conversion, decompression, or indexing) where needed; conversion to mzML is mandatory for early `DIA-NN` versions and for some vendor formats such as SCIEX `.wiff`. A spectral library is then predicted in-silico from the FASTA once (*serial*; skipped when a library is supplied), and every file is calibrated against it in a *parallel* first pass. The first-pass results are merged into a single empirical library (*serial*); each file is searched again against that library *in parallel*; and a final consolidation pass merges all per-file results into the cohort quantification, applying match-between-runs, normalisation, and protein inference (*serial*). MSstats conversion, `pmultiqc` reporting, and QPX export then run once at the end (*serial*). The two per-file search passes carry essentially all of the compute and scale across cluster nodes, driving the throughput reported below (Section 3.2); the serial steps set a largely fixed wall-clock floor. Outputs include the standard `DIA-NN` report tables (`report.tsv`, `report.pr_matrix.tsv`, `report.pg_matrix.tsv`) alongside MSstats-formatted input and a `pmultiqc` [15] quality-control report. A harmonised QPX archive bundles identifications, metadata, and provenance into a Parquet-based artefact, queried directly with DuckDB and, for the quantification layers, exported to MuData (`.h5mu`) for scverse-compatible [17] downstream analysis.

3.2. quantmsdiann throughput and scaling

To characterise how `quantmsdiann` scales on an HPC cluster, we simulated the reanalysis of an entire single-cell experiment of ~2,300 raw files (PXD071075) across cluster sizes from 10 to 300 nodes (set through the Nextflow `executor.queueSize`), and measured the end-to-end wall-clock time (Fig. 1b). Because the per-file work is parallelised across nodes, a single SDRF-driven invocation completed the whole cohort within hours: wall-clock fell monotonically from 37.7 h at 10 nodes to 2.4 h at 300 nodes. The curve has a practical knee near 200 nodes (2.6 h), beyond which the serial cohort-level steps dominate and additional nodes give diminishing returns (a 30× increase in cluster width yields only a 15.5× speed-up), so the cluster should be sized to the desired turn-around time rather than to maximum throughput.

We then compared the time-to-finish across every dataset in this study (Fig. 1c). For proteome-scale cohorts, wall-clock stayed within a few hours irrespective of cohort size, because the per-file passes absorb additional files in parallel: the largest cohort (ProCan, PXD030304; 5,798 TripleTOF 6600 runs) finished in 4.8 h, in the same few-hour band as cohorts orders of magnitude smaller, while the 6-run ProteoBench benchmarks completed in under an hour. Time-to-finish is therefore governed by the serial steps rather than by the number of files: the single longest run is a 10-file phosphoproteomics cohort (PXD049692, 18.4 h), where building the deep variable-modification spectral library, not the per-file search, dominates the wall-clock. Peak memory during the cohort-level consolidation stayed within the 64–128 GB available on commodity HPC nodes, so no high-memory specialist hardware is required.

Per-step wall-clock and resource envelopes are reported in Additional file 1: Figs. S1 and S2.

3.3. ProteoBench benchmark across `DIA-NN` versions and instruments

To test whether `quantmsdiann` preserves `DIA-NN`'s analytical performance under SDRF-driven orchestration, we processed every public ProteoBench DIA module through the workflow and compared the resulting identification counts against the ProteoBench public-submission snapshot (29 May 2026) [10]. Because `quantmsdiann` runs `DIA-NN` library-free (predicting the library in-silico from the FASTA, with no externally supplied experimental library), we restricted the community set for a like-for-like comparison to ProteoBench `DIA-NN` submissions that used the same predicted-from-FASTA library (`predictors_library` = DIANN); submissions that loaded a pre-existing experimental or user-curated spectral library search a different candidate space and are excluded from the headline comparison. The benchmark panel covers four instrument families and acquisition modes: Orbitrap Astral 2 Th DIA (*Module 7*), single-cell DIA on Orbitrap Astral (*Module 9*, PXD049412), timsTOF diaPASEF (*Module 5*, PXD062685), and SCIEX ZenoTOF 7600 (*Module 10*, PXD070049). For each module we ran the workflow against five `DIA-NN` releases (1.8.1, 2.1.0, 2.2.0, 2.3.2, and 2.5.0) in a single invocation, with all five versions parametrised from the same SDRF (Fig. 2).

Precursor counts (quantified in at least one replicate) grew monotonically with `DIA-NN` version on every module, rising by 14% on Module 7 (105,793 $\rightarrow$ 120,973 from 1.8.1 to 2.5.0), 14% on Module 9 (38,218 $\rightarrow$ 43,761), 27% on Module 5 (91,075 $\rightarrow$ 115,735), and 17% on Module 10 (85,847 $\rightarrow$ 100,524). Protein-group counts at 1% FDR increased over the same range (endpoint gains of 7% / 8% / 14% / 7% respectively), though not strictly monotonically: on each module the protein count dips by 1–2% at an intermediate version before the newest release recovers the headline gain. Reproducibility-filtered precursor counts ($\geq$3 replicates) can likewise be non-monotonic: on the ZenoTOF module (PXD070049), for instance, `DIA-NN` 2.5.0 yields the most total precursors but slightly fewer reproducible ones than 2.3.2. This monotonic version-progress trend is consistent with what the broader ProteoBench community reports across `DIA-NN` releases [10], indicating that the SDRF wrapper introduces no version-coupling artefacts. Quantification accuracy, by contrast, is essentially invariant across `DIA-NN` versions. On every module the measured per-species

$\log_2$ fold-changes recover the expected HYE ratios (human at unity, yeast and *E. coli* near their design ratios, with the mild attenuation at the extreme *E. coli* ratio that is characteristic of DIA quantification), and the five `DIA-NN` releases overlie one another at every species (Fig. 2b). The version-to-version shift in median quantification error $|\epsilon|$ ($\sim$0.01 $\log_2$, $\approx$0.5–1% in linear fold-change) is far smaller than the per-precursor spread, so the choice of `DIA-NN` release is best driven by the identification gains in panel (a) rather than by accuracy. On the two modules for which the snapshot contained predicted-library `DIA-NN` submissions, Orbitrap Astral (Module 7, $n = 15$) and timsTOF diaPASEF (Module 5, $n = 8$), `quantmsdiann`'s five versions fall within the community median-$|\epsilon|$ distribution, at or below its median (Fig. 2c); the single-cell and ZenoTOF modules carried no predicted-library community submission. Parameter-matched per-submission comparisons are provided in Additional file 1: Fig. S3.

3.4. Independent reanalyses of public DIA deposits recover comparable or deeper coverage than the original quantifications

To evaluate `quantmsdiann` on production-scale public deposits, we reanalysed independent DIA cohorts, spanning bulk cell lines, single cells, and spatial proteomics, that had been processed and published with different DIA engines (Fig. 3). For each cohort the same raw files were retrieved from PRIDE Archive [6] (or MassIVE, for the single-cell plexDIA cohort), parsed through an SDRF generated from the original sample annotations, and run through `quantmsdiann` with current `DIA-NN` releases.

On the **NCI-60** panel (PXD003539, Guo *et al.* 2019 [18]; originally analysed with OpenSWATH against the Guo assay library), `quantmsdiann` recovered 95,633 peptides at 1% FDR against 40,592 in the original submission (+135%), with comparable protein-group counts (6,747 versus 6,556; Fig. 3a). On the **single-cell oocyte plexDIA** cohort (MSV000093870, Galatidou *et al.* 2024 [19]; mTRAQ 3-channel, Q Exactive), the first plexDIA dataset processed through `quantmsdiann`, the workflow quantified 2,904 protein groups from the same raw data, versus 2,122 in the original analysis (+37%). At the UniProt-accession level it recovered 95% of the originally reported proteins (2,013 of 2,122) and detected 1,539 further accessions, at comparable per-cell median depth (1,736 versus 1,784; Fig. 3b). On the **ProCan**-DepMapSanger 949-cell-line pan-cancer panel (PXD030304, Gonçalves *et al.* 2022 [20]; originally analysed with `DIA-NN` 1.7.x), `quantmsdiann` recovered 8,921 protein groups at 1% FDR against 8,498 originally (+5%) and

7,990 protein groups requiring ≥ 2 unique peptides versus 6,692 (+19%; Fig. 3c). Both original counts are reported in the ProCan paper itself, and the `quantmsdiann` numbers use the matching ProCan-paper filters under our conservative contaminant/decoy policy (Experimental Procedures), so each side of the comparison is like-for-like. A fourth, bulk cohort already analysed with a current DIA engine, the Sun *et al.* 2023 PCT-SWATH breast-cancer panel (PXD004701, 76 lines [21]), yielded near-parity (6,146 versus 6,091 protein groups, +0.9%; Additional file 1), as expected when the original analysis already used a modern engine. On a **spatial** Deep Visual Proteomics cohort of MYCN-amplified neuroblastoma (PXD064049; laser-microdissected regions, diaPASEF, originally analysed with `DIA-NN` 1.8.1), `quantmsdiann` (`DIA-NN` 2.5.0) matched the original at the precursor level (16,443 versus 16,518) and quantified somewhat fewer protein groups (2,425 versus 2,882; Fig. 3d); this protein-group difference reflects `quantmsdiann`'s stricter, entrapment-controlled FDR (empirically 0.7% at the protein-group level) rather than reduced sensitivity. Finally, on a **phosphoproteomics** cohort (PXD049692; NK-cell Fe-NTA enrichment, diaPASEF, originally analysed with Spectronaut directDIA), `quantmsdiann` quantified a comparable number of phosphopeptides at 1% FDR (4,234 versus 4,254; Fig. 3e), matching a field-standard directDIA analysis on phospho-enriched data under the same library-free paradigm. Per-condition completeness, peptide-per-protein distributions, gene-versus-accession overlap and per-cell-line missingness for the cell-line and single-cell cohorts are provided as Additional file 1: Figs. S5–S13.

3.5. Pan-cohort of five public DIA cell-line reanalyses

To illustrate the value of running the same SDRF-driven workflow across many independent deposits, we combined the three reanalysed cohorts above with two additional cell-line DIA panels (PXD041421, the MultiPro batch-effect testbed, Wang *et al.* 2023 [22]; PXD017199, Tognetti *et al.* 2021 [23], 67 breast-cancer lines) into a single `quantmsdiann` invocation and assembled a pan-cohort view of per-cohort yield and per-tissue coverage (Fig. 4). Per-cohort headline counts are summarised in Fig. 4a. Across the five cohorts the reanalysis spans 12,229 unique UniProt accessions, of which 5,144 (42.1%) are detected in all five (a pan-cohort core proteome) and a further 1,334 (10.9%) in exactly four; these overlap counts are tabulated in the combined per-cohort counts provided with the analysis code.

Tissue-level coverage across the combined panel spans 27 Cellosaurus-derived tissue categories (plus the catch-all "Other tissue" and "Healthy (Noncancer)" bins; Fig. 4b,c): lung, breast, haematopoietic-and-lymphoid, skin, and central nervous system together account for 59% of all cell-line entries in the pan-cohort. Because every count comes from a single SDRF-parameterised `quantmsdiann` invocation, the resulting protein matrix is queryable from the published QPX archive and can be filtered, joined, or re-grouped by Cellosaurus, NCIt, or Disease Ontology terms without manual harmonisation. The consolidated SDRF and per-cohort `quantmsdiann` configurations underpinning this pan-cohort run will be deposited on Zenodo before publication, so that the full input–configuration–output triple can be re-executed by third parties.

4. Discussion

`quantmsdiann` automates the path from a deposited DIA dataset to a reproducible quantification table, at a scale that supports systematic reanalysis of public DIA archives. The SDRF-first design encourages experiment annotation upstream of analysis and produces metadata that is interoperable with the wider proteomics ecosystem [8]; the QPX archive bundles the SDRF, harmonised quantification tables and pipeline provenance into a single Parquet container so that reanalysed results can be archived (e.g. on Zenodo) and re-queried alongside the original raw data. By focusing on `DIA-NN` rather than the multi-engine scope of the related bigbio/quantms pipeline [7], `quantmsdiann` achieves a leaner dependency surface and faster pipeline-level execution for the DIA-only regime, where the dominant cost is per-file `DIA-NN` invocation rather than across-engine comparison.

This parallel structure also shapes how the workflow scales. End-to-end wall-clock is the sum of a largely fixed serial overhead (chiefly in-silico spectral-library generation from the FASTA, the single largest non-parallel step, together with cohort-level consolidation) and the per-file analysis stages, which run as independent tasks across the cluster (Fig. 1b) and whose elapsed time grows only sub-linearly with cohort size as files are processed in parallel waves (per-step envelopes in Additional file 1: Fig. S1). Time-to-finish is therefore dominated by the fixed overhead for small cohorts and stays within a few hours even for the largest deposits, so a small dataset is not necessarily faster than a much larger one (Fig. 1c); the serial overhead itself scales with the predicted-library size, growing for wide variable-modification searches such as phosphoproteomics, rather than with the number of files.

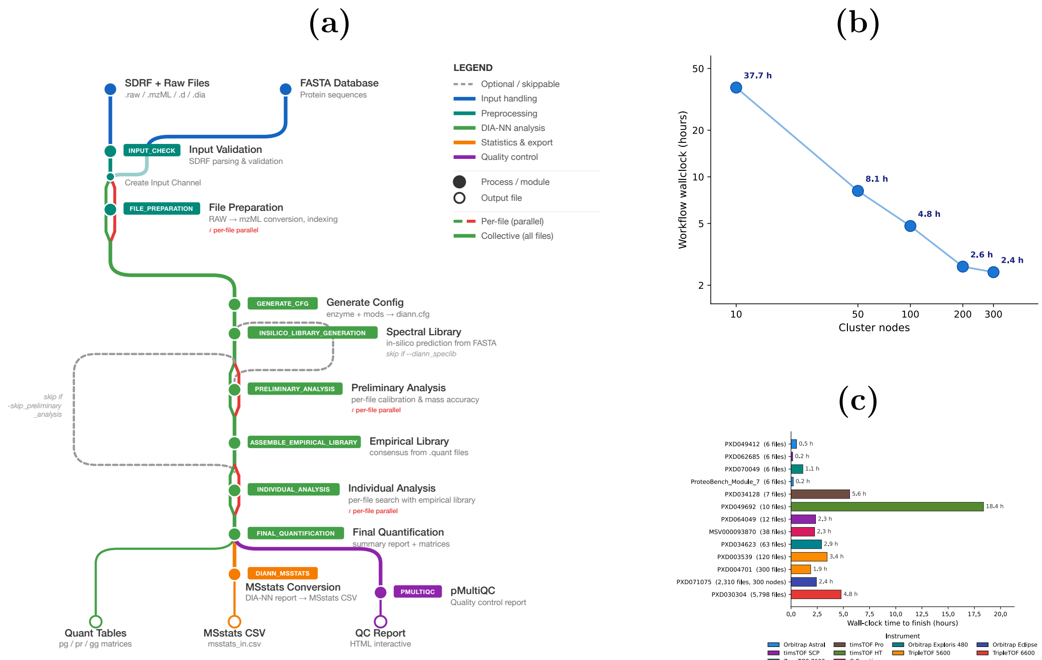


Figure 1: `quantmsdiann` **pipeline architecture and runtime scaling.** (**a**) Subway diagram of the `quantmsdiann` workflow: mandatory modules as filled circles, optional/skippable modules as open circles. Per-file parallel stages (red; file preparation, preliminary calibration, individual search) run as one task per raw file, whereas collective stages (green; in-silico library generation, empirical library assembly, final quantification, MSstats conversion, and `pmultiqc` reporting) run once over the full cohort. (**b**) Workflow wall-clock time versus the number of cluster nodes (Nextflow `executor.queueSize`) on the PXD071075 single-cell cohort. (**c**) Wall-clock time to finish each reanalysis, one bar per dataset, ordered by cohort size (number of MS data files) and coloured by instrument family. For proteome-scale cohorts, time-to-finish stays within a few hours regardless of file count because per-file work is parallelised; the longest run is a phosphoproteomics cohort (PXD049692), where generating the deep variable-modification spectral library dominates the wall-clock.

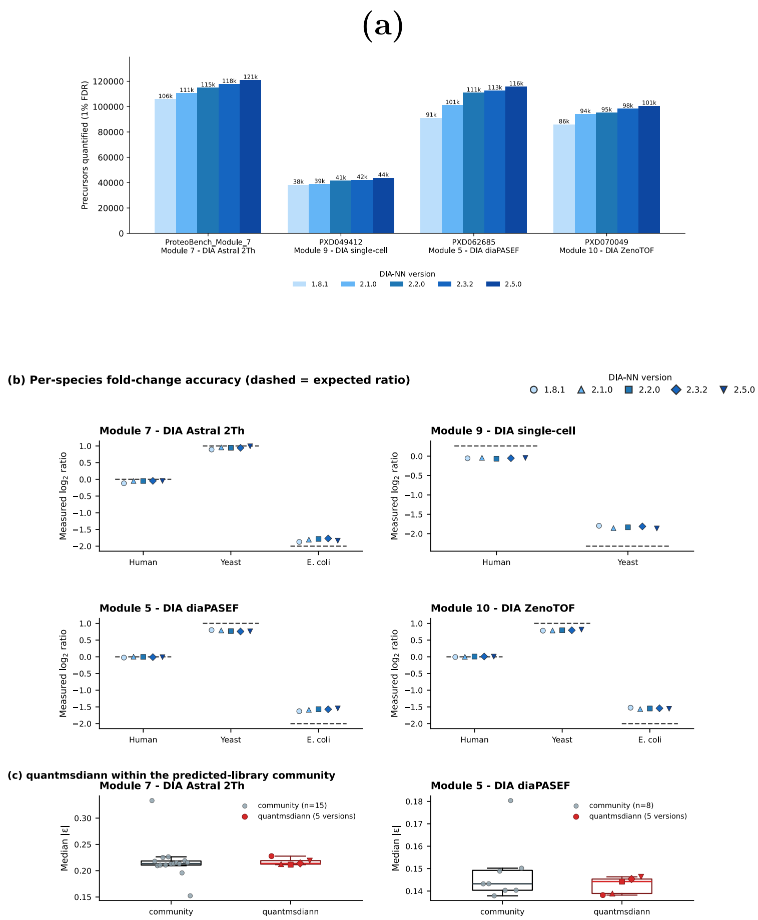
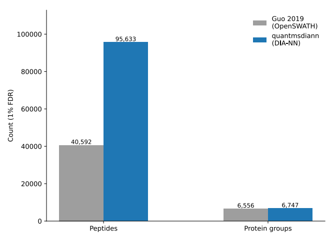
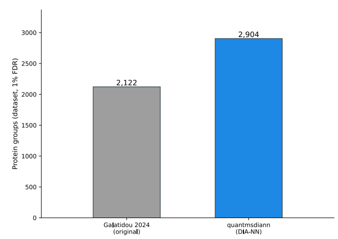


Figure 2: **`quantmsdiann` on ProteoBench across five `DIA-NN` releases and four DIA modalities** (Orbitrap Astral, Module 7; single-cell, Module 9/PXD049412; timsTOF dia-PASEF, Module 5/PXD062685; SCIEX ZenoTOF, Module 10/PXD070049). (**a**) Precursors at 1% FDR by `DIA-NN` version (1.8.1–2.5.0; light to dark blue) from one `quantmsdiann` invocation. (**b**) Per-species quantification accuracy: measured $\log_2$ fold-change for each HYE species against the ProteoBench-expected ratio (dashed), with all five `DIA-NN` versions overlaid (light to dark). (**c**) For the two modules with predicted-from-FASTA community submissions, `quantmsdiann`'s five versions (red) against the community median-$|\epsilon|$ distribution (grey; box = IQR). See Additional file 1: Fig. S3 for parameter-matched per-submission comparisons.

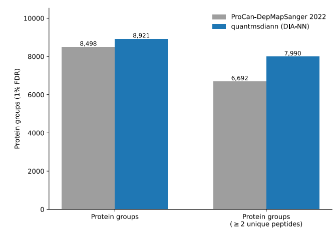
(a) PXD003539 (NCI-60, 2019)



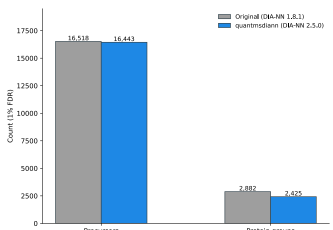
(b) MSV000093870 (plexDIA, 2024)



(c) PXD030304 (ProCan, 2022)



(d) PXD064049 (MYCN DVP spatial, 2025)



(e) PXD049692 (NK phospho, 2024)

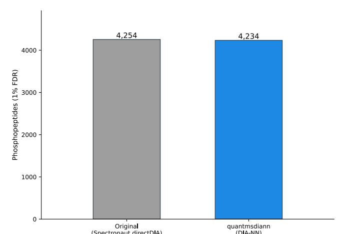


Figure 3: **Independent reanalysis of public DIA datasets, spanning bulk cell lines, single cells, spatial proteomics, and phosphoproteomics, recovers comparable or greater identification depth than the original deposits.** Each panel compares the original published counts (grey) with the `quantmsdiann` reanalysis at 1% FDR (blue) from the same raw data. (**a**) NCI-60 bulk cell lines (PXD003539, Guo 2019; original OpenSWATH analysis). (**b**) Single-cell oocyte plexDIA (MSV000093870, Galatidou 2024). (**c**) ProCan-DepMapSanger pan-cancer cell-line panel (PXD030304). (**d**) MYCN-amplified neuroblastoma Deep Visual Proteomics (spatial, laser-microdissected regions; PXD064049, diaPASEF), with precursors and protein groups compared against the original DIA-NN 1.8.1 analysis. (**e**) NK-cell Fe-NTA phosphoproteomics (PXD049692, diaPASEF): phosphopeptides compared against the original Spectronaut directDIA analysis on the same runs. Per-cohort detail for panels (a)–(c) in Additional file 1: Figs. S5–S13.

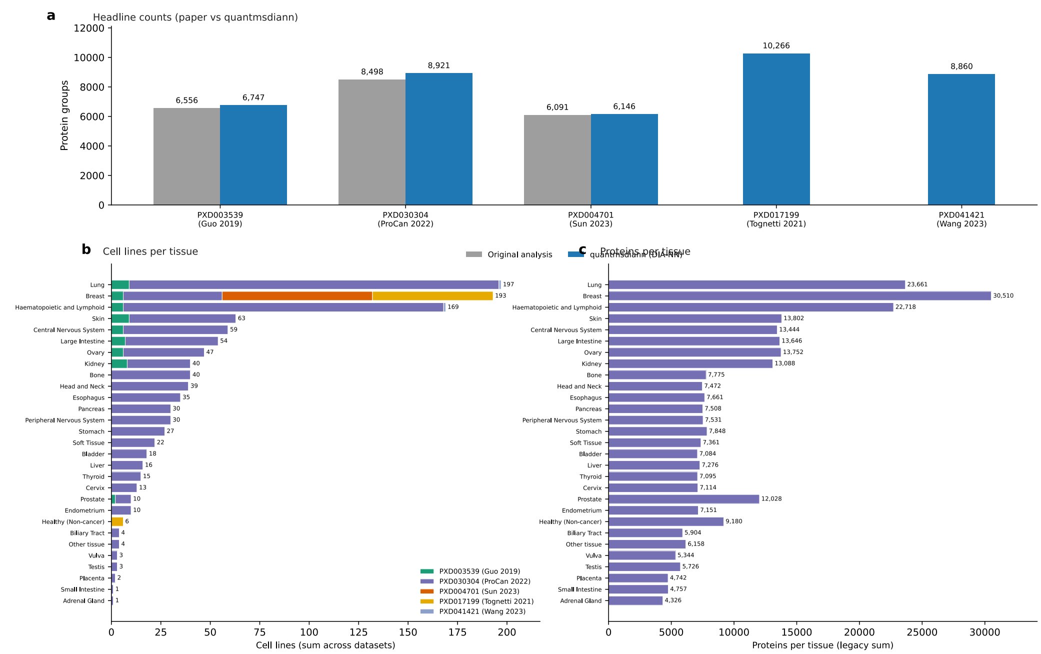


Figure 4: **Pan-cohort of five public DIA cell-line reanalyses by quantmsdiann.** Combines PXD003539 (NCI-60, Guo 2019), PXD030304 (ProCan-DepMapSanger, 949 lines), PXD004701 (Sun 2023, 76 breast cancer lines), PXD041421 (Wang 2023), and PXD017199 (Tognetti 2021, 67 breast cancer lines), reanalysed under a single `quantmsdiann` invocation. (**a**) Per-cohort headline protein-group counts: the originally published count (grey) and the `quantmsdiann` reanalysis (blue); PXD017199 and PXD041421 have no published DIA headline, so only the `quantmsdiann` bar is shown. (**b**) Cell lines per pan-cancer tissue, stacked by contributing cohort. (**c**) Unique UniProt proteins detected per tissue across the combined panel.

The demonstrations stress the workflow along three orthogonal axes: analytical fidelity (ProteoBench; Fig. 2), recovery against published per-study quantifications (independent reanalyses; Fig. 3), and archive-scale throughput (pan-cohort; Fig. 4). The recovery results suggest that the value of DIA reanalysis is real but time-limited (largest for deposits originally analysed with older or non-DIA-NN engines and shrinking toward parity for recently processed cohorts), a regime an SDRF-driven workflow is suited to address as new DIA-NN versions ship. The workflow already supports broader single-cell and laser-microdissected spatial acquisition modes (Fig. 1a, library-handling branch), and SDRFs for the Brunner [2], Derks [24], Leduc [25], Wang [4], Mund [26] and Makhmut [27] datasets are in preparation.

The cross-DIA-NN-version reproducibility view enabled by quantmsdiann is, to our knowledge, the first pipeline-level treatment of this question for DIA-NN. We expect this to become increasingly important as DIA-NN evolves: a deposited single-cell DIA dataset analysed with DIA-NN 1.8 today and re-analysed with DIA-NN 2.x in two years should not produce different biological conclusions for the same data and parameters. In current practice such re-analyses are rarely performed, because the bespoke per-study scripts make them prohibitively expensive, a barrier the single-invocation audit of Section 3.3 removes.

quantmsdiann has several limitations. First, quantmsdiann does not currently include batch correction, cell-type assignment, or differential-abundance modules; these are best handled in downstream R or Python packages such as scp [28], which can read the QPX or MSstats outputs directly. Second, library generation for plexDIA channel deconvolution remains internal to DIA-NN, and quantmsdiann inherits any limitations of that step. Third, the QPX format will continue to evolve as community feedback accumulates; backward compatibility is maintained through schema versioning. Fourth, the single-cell, spatial, and phosphoproteomics demonstrations here (the oocyte plexDIA, the MYCN-amplified neuroblastoma Deep Visual Proteomics, and the NK-cell phospho cohorts) are proof-of-principle; broader benchmarking across these modalities (Fig. 1a, library-handling branch) remains future work.

Finally, we note that quantmsdiann is complementary to, rather than competing with, existing single-cell DIA workflows. The benchmarking work of Wang *et al.* [4] compared multiple search engines and rescoring strategies; quantmsdiann adopts the broadly competitive DIA-NN engine and focuses on the orthogonal problem of orchestrating that engine reproducibly at scale. Similarly, earlier nf-core DIA efforts such as DIAproteomics [29] and, more recently, the MHCquant2 pipeline [30] address analogous orchestration problems for DIA and immunopeptidomics. Together, these contributions extend the nf-core proteomics ecosystem to cover the major MS-based proteomics modalities behind a common SDRF-driven interface.

Data Availability

All datasets reanalysed in this study are publicly available through ProteoMeXchange. Cell-line bulk DIA reanalyses: PRIDE PXD003539 (Guo 2019, NCI-60), PXD004701 (Sun 2023), PXD030304 (ProCan-DepMapSanger), PXD017199 (Tognetti 2021), PXD041421 (Wang 2023). Spatial Deep Visual Proteomics reanalysis: PRIDE PXD064049 (MYCN-amplified neuroblastoma, diaPASEF). Scalability experiment: PRIDE PXD071075. ProteoBench benchmarks: PXD049412 (single-cell, Module 9), PXD062685 (diaPASEF, Module 5), PXD070049 (ZenoTOF, Module 10), and Module 7 (Astral) datasets. The full list of PRIDE accessions used in the pan-cohort of DIA cell-line reanalyses is provided in Additional file 1: Table S1. `quantmsdiann` is available at `https://github.com/bigbio/quantmsdiann` under the MIT license; documentation is hosted at `https://quantmsdiann.quantms.org/`. The pipeline is implemented in Nextflow DSL2 ($\geq$25.10.4) and packages each step as a versioned Docker, Singularity, or Apptainer container image. `DIA-NN` 1.8.1 is distributed publicly via BioContainers; because the `DIA-NN` license restricts redistribution of later releases, containers for `DIA-NN` 1.9 and above are built locally from the recipes in `https://github.com/bigbio/quantms-containers` (see Experimental Procedures). Pipeline version v2.1.0 (release codename "Sao Pablo", 2026-05-08) was used for the analyses reported in this paper. Analysis scripts that generated the figures in this manuscript are available at `https://github.com/bigbio/quantmsdiann-paper` (this repository).

Supplemental Data

This article contains supplemental data. Additional file 1 provides Table S1 (the full PRIDE accession list), per-process runtime and resource envelopes (Figs. S1–S2), ProteoBench supplementary panels (Figs. S3–S4), and per-cohort completeness, peptide-per-protein and gene-versus-accession overlap figures (Figs. S5–S13).

Conflict of Interest

The authors declare no competing interests.

Acknowledgments

[We thank the Brunner, Derks, Leduc, Mund, Wang, and Coscia teams for depositing their data publicly. We thank the nf-core community and the OpenMS and DIA-NN developer teams for their ongoing work.] M.B. is supported by the Science and Technology Innovation Key R&D Program of Chongqing (CSTB2023TIAD-STX0002). Y.P-R. acknowledges support from the Wellcome Trust [223745/Z/21/Z], Open Targets [OTAR03-91], BBSRC [BB/V018779/1, BB/X001911/1], EPSRC [EP/Y035984/1], and EMBL core funding. A.L. acknowledges support from MCIN (Recovery, Transformation and Resilience Plan); the Basque Government "Biotechnology Complementary Plan Applied to Health" funded by the European Union NextGeneration EU (PRTR-C17.I1; PRTR-C17.I01.P01.S13) (AAAA_ACG_AY_2539/22_05); and an EMBO Scientific Exchange Grant (number 10015/2023). *[Funding sources and grant numbers for the remaining authors.]*

Author Contributions

QXY, YS and CD developed the pipeline. HW helped build the containers and contributed to the workflow implementation and manuscript writing. AL annotated the SDRF metadata and helped run the workflow. TS, FC, NS and JN assisted with the DIA-NN reanalyses and manuscript writing; TS additionally contributed to the workflow design, and JN tested the tool and identified multiple issues. OK provided feedback on the plexDIA datasets. VD provided DIA-NN expertise and guidance on cross-version reproducibility. MB contributed to manuscript writing and supervised the students; MB and YPR conceived and supervised the project. QXY and YPR wrote the original draft, with input from all authors. All authors read and approved the final manuscript.

Abbreviations

CLI	command-line interface
DDA	data-dependent acquisition
DIA	data-independent acquisition

DIA-NN data-independent acquisition neural networks (DIA search and quantification software)

diaPASEF data-independent acquisition with parallel accumulation–serial fragmentation

FDR false discovery rate

HPC high-performance computing

HYE human, yeast, and *E. coli* (ProteoBench three-proteome mixture)

IQR interquartile range

LSF Load Sharing Facility (HPC scheduler)

MS mass spectrometry

MS1/MS2 first-/second-stage mass spectrometry (precursor/fragment scans)

PASEF parallel accumulation–serial fragmentation

QPX Quantitative Proteomics eXchange

SDRF Sample and Data Relationship Format

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